OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

14.8 Signal semantics

Signals are uniquely defined by the @country, @year, @type, and @subtype attributes. These attributes specify the visualization of the signals because the visualization is different for each country. However, the behavior of the traffic participant is often identical independent of the country, for example, for a stop sign. A simulation engine requires rules specified for all signs for all countries.

The introduction of signal semantics reduces the effort to specify the rules for all signs for all countries. For signal semantics the focus is on the behavior of traffic. Functionality for signal semantics cannot be used for visualization.

As traffic behavior is very complex and also depending on other ASAM OpenDRIVE definitions, for example, the lane width, junctions, and road mark, signal semantics are reduced to a very limited scope. The scope of signal semantics in ASAM OpenDRIVE is limited to traffic behavior that is specified just by signals in ASAM OpenDRIVE. Each traffic behavior is specified by a specific element, for example, by the `<priority>` element that specifies priority regulations.

Table 133. Possible Semantics

Traffic behavior	Description	Example	Link to attribute	Usage in/for `<defaultRegulations>` in `<header>`	Usage in/for `<signal>` or `<sign>`
<lane>	Specifies lane regulations.	Sign that defines cars are not allowed to overtake.	`<lane>`	not recommended	recommended
<parking>	Specifies parking regulations.	Sign that defines that parking is forbidden.	no attributes	not allowed	recommended
<priority>	Specifies priority regulations.	Stop sign	`<priority>`	recommend for default junction priority	recommended

Traffic behavior	Description	Example	Link to attribute	Usage in/for `<defaultRegulations>` in `<header>`	Usage in/for `<signal>` or `<sign>`
<prohibited>	Specifies that certain types of traffic participants are not allowed to enter. Signal semantics for traffic participants in ASAM OpenDRIVE are currently not defined because traffic participants are not harmonized for all standards.	No pedestrians sign	no attributes	recommend for default junction priority	recommended
<routing>	Specifies routing information.	London	no attributes	not allowed	recommended
<speed>	Specifies speed regulations.	Speed limit sign	`<speed>`	recommended	recommended
<streetname>	Specifies the name of a street.	High street	no attributes	not allowed	recommended
<tourist>	Specifies tourist information.	Science museum	no attributes	not allowed	recommended
<warning>	Specifies warnings for traffic participant.	Deer might cross the road	no attributes	not allowed	recommended

Traffic behavior	Description	Example	Link to attribute	Usage in/for `<defaultRegulations>` in `<header>`	Usage in/for `<signal>` or `<sign>`
<supplementaryAllows>	This signal semantic has no meaning on its own. It specifies the type of the traffic participant an exception is made for. Signal semantics for traffic participants in ASAM OpenDRIVE are currently not specified because traffic participants are not harmonized for all standards.	Except bicycles sign	no attributes	not allowed	recommended
<supplementaryDistance>	This signal semantic has no meaning on its own. It specifies the distance after a sign becomes valid or the range in which the sign is valid.	200m sign	`<supplementaryDistance>`	not allowed	recommended
<supplementaryEnvironment>	This signal semantic has no meaning on its own. It specifies under which environmental conditions a sign is valid.	Rain sign	`<supplementaryEnvironment>`	not allowed	recommended

Traffic behavior	Description	Example	Link to attribute	Usage in/for `<defaultRegulations>` in `<header>`	Usage in/for `<signal>` or `<sign>`
<supplementaryExplanatory>	This signal semantic has no meaning on its own. It specifies explanations for a sign.	Emission reduction	no attributes	not allowed	recommended
<supplementaryProhibits>	This signal semantic has no meaning on its own. It specifies the type of the traffic participant a restriction is made for. Signal semantics for traffic participants in ASAM OpenDRIVE are currently not specified because traffic participants are not harmonized for all standards.	Truck Symbol	no attributes	not allowed	recommended
<supplementaryTime>	This signal semantic has no meaning on its own. It specifies the time or date a sign is valid.	9:00 - 18:00	`<supplementaryTime>`	not allowed	recommended

Elements in UML model

`<semantics>` element

```
UML class:  t_signals_semantics
XML tag:    <semantics> (Multiplicity: 0..1)
Introduced: 1.8.0
```

Semantics are limited to traffic behavior that is specified just by signals in ASAM OpenDRIVE. Each traffic behavior is specified by a specific element.

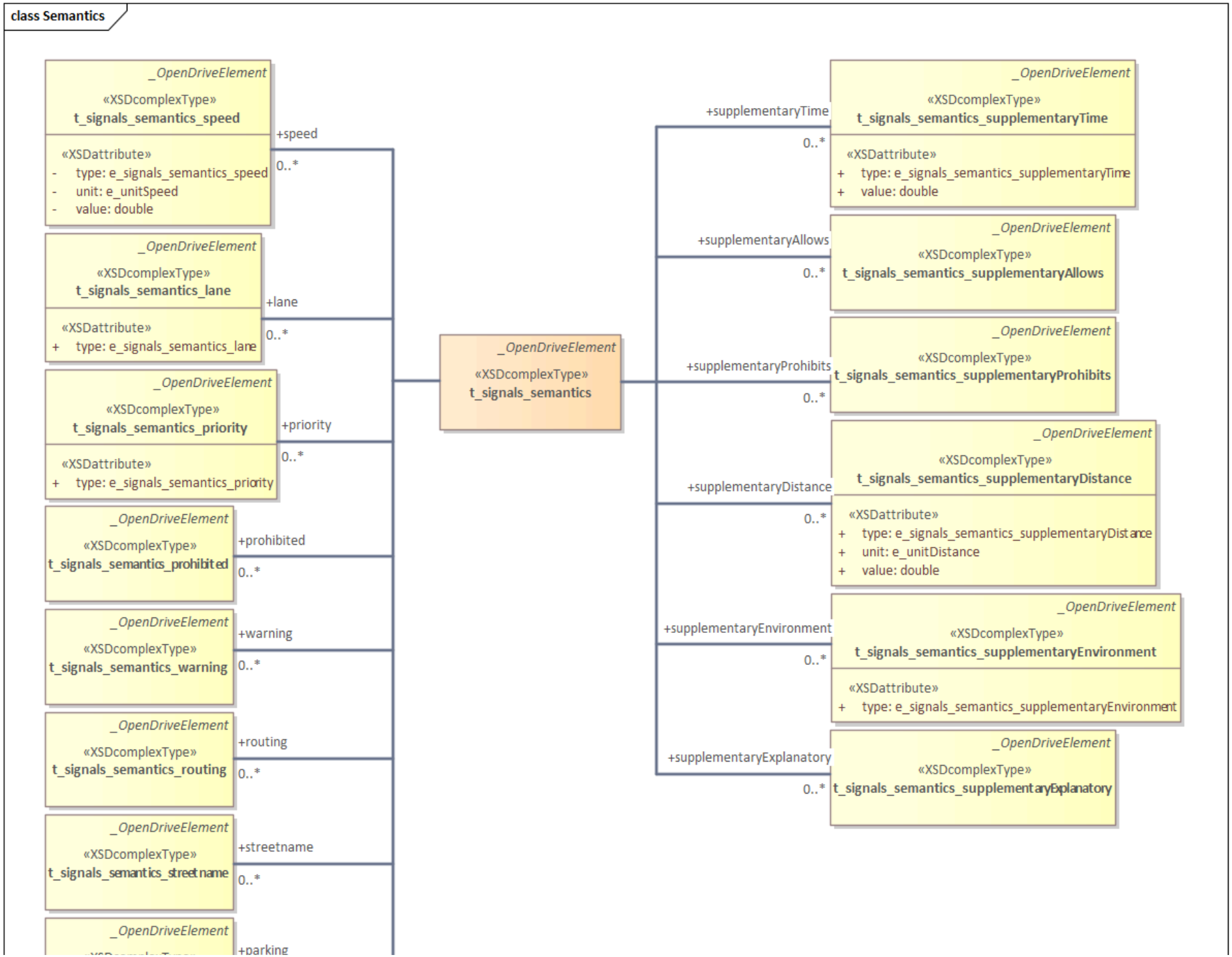
class Semantics
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_speed
«XSDattribute»
- type: e_signals_semantics_speed
- unit: e_unitSpeed
- value: double
+speed
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_lane
«XSDattribute»
+ type: e_signals_semantics_lane
+lane
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_priority
«XSDattribute»
+ type: e_signals_semantics_priority
+priority
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_prohibited
+prohibited
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_warning
+warning
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_routing
+routing
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_streetname
+streetname
0..*
_OpenDriveElement
+parking
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics
+supplementaryTime
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_supplementaryTime
«XSDattribute»
+ type: e_signals_semantics_supplementaryTime
+ value: double
+supplementaryAllows
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_supplementaryAllows
+supplementaryProhibits
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_supplementaryProhibits
+supplementaryDistance
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_supplementaryDistance
«XSDattribute»
+ type: e_signals_semantics_supplementaryDistance
+ unit: e_unitDistance
+ value: double
+supplementaryEnvironment
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_supplementaryEnvironment
«XSDattribute»
+ type: e_signals_semantics_supplementaryEnvironment
+supplementaryExplanatory
0..*
_OpenDriveElement
«XSDcomplexType»
t_signals_semantics_supplementaryExplanatory

Figure 136. UML class diagram of the Semantics class

Figure 136 shows the UML class diagram of the ASAM OpenDRIVE Semantics class.

`<lane>` element

```
UML class:  t_signals_semantics_lane
XML tag:    <lane> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Specifies lane regulations.

Table 134. Attributes of the <lane> element

Name	Type	Use	Introduced
`type`	e_signals_semantics_lane	required	1.8.0

`<parking>` element

```
UML class:  t_signals_semantics_parking
XML tag:    <parking> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Specifies parking regulations.

`<priority>` element

```
UML class:  t_signals_semantics_priority
XML tag:    <priority> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Specifies priority regulations.

Table 135. Attributes of the <priority> element

Name	Type	Use	Introduced
`type`	e_signals_semantics_priority	required	1.8.0

`<prohibited>` element

```
UML class:  t_signals_semantics_prohibited
XML tag:    <prohibited> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Specifies that certain types of traffic participants are not allowed to enter. Signal semantics for traffic participants in ASAM OpenDRIVE are currently not defined because traffic participants are not harmonized for all standards.

`<routing>` element

```
UML class:  t_signals_semantics_routing
XML tag:    <routing> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Specifies routing information.

`<speed>` element

```
UML class:  t_signals_semantics_speed
XML tag:    <speed> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Specifies speed regulations.

Table 136. Attributes of the <speed> element

Name	Type	Use	Introduced
`type`	e_signals_semantics_speed	required	1.8.0
`unit`	e_unitSpeed	required	
`value`	double	required	

`<streetname>` element

```
UML class:  t_signals_semantics_streetname
XML tag:    <streetname> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Specifies the name of a street.

`<tourist>` element

```
UML class:  t_signals_semantics_tourist
XML tag:    <tourist> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Specifies tourist information.

`<warning>` element

```
UML class:  t_signals_semantics_warning
XML tag:    <warning> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Specifies warnings for traffic participant.

`<supplementaryAllows>` element

```
UML class:  t_signals_semantics_supplementaryAllows
XML tag:    <supplementaryAllows> (Multiplicity: 0..*)
Introduced: 1.8.0
```

This signal semantic has no meaning on its own. It specifies the type of the traffic participant an exception is made for. Signal semantics for traffic participants in ASAM OpenDRIVE are currently not specified because traffic participants are not harmonized for all standards.

`<supplementaryDistance>` element

```
UML class:  t_signals_semantics_supplementaryDistance
XML tag:    <supplementaryDistance> (Multiplicity: 0..*)
Introduced: 1.8.0
```

This signal semantic has no meaning on its own. It specifies the distance after a sign becomes valid or the range in which the sign is valid.

Table 137. Attributes of the <supplementaryDistance> element

Name	Type	Use	Introduced
`type`	e_signals_semantics_supplementaryDistance	required	1.8.0
`unit`	e_unitDistance	required	1.8.0
`value`	double	required	1.8.0

`<supplementaryEnvironment>` element

```
UML class:  t_signals_semantics_supplementaryEnvironment
XML tag:    <supplementaryEnvironment> (Multiplicity: 0..*)
Introduced: 1.8.0
```

This signal semantic has no meaning on its own. It specifies under which environmental conditions a sign is valid.

Table 138. Attributes of the <supplementaryEnvironment> element

Name	Type	Use	Introduced
`type`	e_signals_semantics_supplementaryEnvironment	required	1.8.0

`<supplementaryExplanatory>` element

```
UML class:  t_signals_semantics_supplementaryExplanatory
XML tag:    <supplementaryExplanatory> (Multiplicity: 0..*)
Introduced: 1.8.0
```

This signal semantic has no meaning on its own. It specifies explanations for a sign.

`<supplementaryProhibits>` element

```
UML class:  t_signals_semantics_supplementaryProhibits
XML tag:    <supplementaryProhibits> (Multiplicity: 0..*)
Introduced: 1.8.0
```

This signal semantic has no meaning on its own. It specifies the type of the traffic participant a restriction is made for. Signal semantics for traffic participants in ASAM OpenDRIVE are currently not specified because traffic participants are not harmonized for all standards.

`<supplementaryTime>` element

```
UML class:  t_signals_semantics_supplementaryTime
XML tag:    <supplementaryTime> (Multiplicity: 0..*)
Introduced: 1.8.0
```

This signal semantic has no meaning on its own. It specifies the time or date a sign is valid.

Table 139. Attributes of the <supplementaryTime> element

Name	Type	Use	Introduced
`type`	e_signals_semantics_supplementaryTime	required	1.8.0
`value`	double	required	1.8.0

XML example

```xml
<signal s="4.0"
        t="1.0"
        id="1"
        name="SpeedLimit60"
        dynamic="no"
        orientation="+"
        zOffset="2.00"
        country="DE"
        countryRevision="2013"
        type="274"
        subtype="56"
        value="60"
        unit="km/h"
        hOffset="0"
        pitch="0"
        roll="0"
        height="0.76"
        width="0.76">
    <semantics>
        <speed type="maximum" value="60" unit="km/h"/>
    </semantics>
</signal>
```

Rules

- Each `<signal>` element may have one ore more semantic elements.
- Signal semantics shall not be specified for signs if no category for the desired traffic behavior exists.

Related topics

- Section 6.4.3, " `<defaultRegulations>` element"